CHAPTER 4

Series Solution of Differential Equations

Chapter Outline

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4.1 INTRODUCTION

In general, the solutions to differential equations with variable coefficients cannot be expressed as a finite linear combination of known elementary functions. In such cases, the solution can be obtained in the form of an infinite convergent series. In this chapter, the power-series method and an extension of the power-series method, called the *Frobenius method*, are used to solve differential equations with variable coefficients.

4.2 POWER-SERIES METHOD

Consider a homogeneous linear second-order differential equation with variable coefficients

$$P_0(x)\frac{d^2y}{dx^2} + P_1(x)\frac{dy}{dx} + P_2(x)y = 0 \qquad \ldots(4.1)$$

where P_0, P_1, and P_2 are polynomials in x.
Dividing Eq. (4.1) by $P_0(x)$,

$$\frac{d^2y}{dx^2} + \frac{P_1(x)}{P_0(x)}\frac{dy}{dx} + \frac{P_2(x)}{P_0(x)}y = 0$$

$$\frac{d^2y}{dx^2} + P(x)\frac{dy}{dx} + Q(x)y = 0 \qquad \ldots(4.2)$$

where $P(x) = \dfrac{P_1(x)}{P_0(x)}$ and $Q(x) = \dfrac{P_2(x)}{P_0(x)}$

Equation (4.2) is known as the standard form (normal form or canonical form) of Eq. (4.1).

The power-series solution of Eq. (4.2) about a point $x = x_0$ depends on the following definitions.

Ordinary Point

A point x_0 is called an ordinary point of Eq. (4.2) if $P(x)$ and $Q(x)$ are both analytic (i.e., differentiable) at x_0.

Notes:

(i) If $P_0(x) \neq 0$ at $x = x_0$ then x_0 is an ordinary point.

(ii) The ordinary point is also known as a regular point of the equation.

Singular Point

A point x_0 is called a singular point of Eq. (4.2) if either $P(x)$ or $Q(x)$ or both are not analytic at x_0.

Note: If $P_0(x) = 0$ at $x = x_0$ then x_0 is a singular point.

Regular Singular Point

A singular point x_0 is called a regular singular point of Eq. (4.2) if $(x - x_0)\,P(x)$ and $(x - x_0)^2\,Q(x)$ both are analytic (i.e., differentiable) at x_0.

Irregular Singular Point

A singular point x_0 is called an irregular singular point of Eq. (4.2) if either $(x - x_0)\,P(x)$ or $(x - x_0)^2\,Q(x)$ or both are not analytic at x_0.

Example 1

Classify the singular points of the differential equation $x^2y'' + xy' - 2y = 0$.

Solution

$$x^2y'' + xy' - 2y = 0 \qquad \ldots(1)$$

$$P_0(x) = x^2$$

At singular points,

$$P_0(x) = 0$$

$$x^2 = 0$$

$$x = 0$$

Hence, $x = 0$ is a singular point.

Dividing Eq. (1) by x^2,

$$y'' + \frac{1}{x}y' - \frac{2}{x^2}y = 0$$

Comparing with the standard form,

$$y'' + P(x)y' + Q(x)y = 0$$

$$P(x) = \frac{1}{x}, \qquad Q(x) = -\frac{2}{x^2}$$

$$xP(x) = 1, \qquad x^2Q(x) = -2$$

Since $xP(x)$ and $x^2Q(x)$ are both analytic (i.e., differentiable) at $x = 0$, it is a regular singular point of Eq. (1).

Example 2

Find the ordinary points and singular points of the equation $(1-x)^2y'' - 6xy' - 4y = 0$.

Solution

$$(1-x)^2y'' - 6xy' - 4y = 0 \qquad \ldots(1)$$

$$P_0(x) = 1 - x^2$$

At singular points,

$$P_0(x) = 0$$

$$1 - x^2 = 0$$

$$x = \pm 1$$

Hence, $x = \pm 1$ are singular points of Eq. (1).

Dividing Eq. (1) by $(1-x^2)$,

$$y'' - \frac{6x}{(1-x^2)}y' - \frac{4}{1-x^2}y = 0$$

Comparing with the standard form,

$$y'' + P(x)y' + Q(x) = 0$$

$$P(x) = -\frac{6x}{1-x^2}, \qquad Q(x) = -\frac{4}{(1-x^2)}$$

$$= \frac{6x}{(x+1)(x-1)}, \qquad = \frac{4}{(x+1)(x-1)}$$

(i) For $x = 1$,

$$(x-1)P(x) = \frac{6x}{x+1}, \qquad (x-1)^2Q(x) = \frac{4(x-1)}{x+1}$$

Since $(x-1)P(x)$ and $(x-1)^2Q(x)$ are both analytic (i.e., differentiable) at $x = 1$, it is a regular singular point.

(ii) For $x = -1$,

$$(x+1)P(x) = \frac{6x}{x-1}, \qquad (x+1)^2 Q(x) = \frac{4(x+1)}{x-1}$$

Since $(x + 1)\,P(x)$ and $(x + 1)^2\,Q(x)$ are both analytic (i.e., differentiable) at $x = -1$, it is a regular singular point of the equation.
All the values of $x \neq \pm 1$ are the ordinary points of the equation.

Example 3

Classify the singular points of the equation

$$x^3(x-2)y'' + x^3y' + 6y = 0$$

Solution

$$x^3(x-2)y'' + x^3y' + 6y = 0 \qquad \text{...(1)}$$

$$P_0(x) = x^3\,(x-2)$$

At singular points,

$$P_0(x) = 0$$

$$x^3(x-2) = 0$$

$$x = 0, \qquad x = 2$$

Hence, $x = 0$ and $x = 2$ are singular points of Eq. (1).
Dividing Eq. (1) by $x^3(x-2)$,

$$y'' + \frac{1}{x-2}y' + \frac{6}{x^3(x-2)}y = 0$$

Comparing with the standard form,

$$y'' + P(x)y' + Q(x) = 0$$

$$P(x) = \frac{1}{x-2}, \qquad Q(x) = \frac{6}{x^3(x-2)}$$

(i) For $x = 0$,

$$xP(x) = \frac{x}{x-2}, \qquad x^2Q(x) = \frac{6}{x(x-2)}$$

Since $x^2Q(x)$ is not analytic (i.e., not differentiable) at $x = 0$, it is an irregular singular point of the equation.

(ii) For $x = 2$,

$$(x-2)P(x) = 1, \qquad (x-2)^2Q(x) = \frac{6(x-2)}{x^3}$$

Since $(x - 2)\,P(x)$ and $(x - 2)^2\,Q(x)$ are both analytic (i.e., differentiable) at $x = 2$, it is a regular singular point of the equation.

EXERCISE 4.1

Find ordinary points and singular points of the following differential equations. Also, classify the singular points.

1. $x^2y'' - 5y' + 3x^2y = 0$

[**Ans.** : $x = 0$, irregular singular point]

2. $e^x y'' + y' - xy = 0$

[**Ans.** : All values of x are ordinary points]

3. $(x^2 + x - 2)^2 y'' + 3(x + 2)y' + (x - 1)y = 0$

[**Ans.** : $x = -2$, regular singular point,
$x = 1$, irregular singular point]

4. $x^3y'' + 3xy' + 6y = 0$

[**Ans.** : $x = 0$, irregular singular point]

5. $x^2y'' + (\sin x)y' + (\cos x)y = 0$

[**Ans.** : $x = 0$, regular singular point]

4.3 SERIES SOLUTION ABOUT AN ORDINARY POINT

Let the power-series solution of the equation

$$P_0(x)y'' + P_1(x)y' + P_2(x)y = 0 \qquad ...(4.3)$$

about an ordinary point x_0 be given as

$$y = \sum_{n=0}^{\infty} a_n (x - x_0)^n = a_0 + a_1(x - x_0) + a_2(x - x_0)^2 + \cdots$$

The coefficients $a_1, a_2, a_3, \ldots$ are obtained as follows:

(i) Let $y = \sum_{n=0}^{\infty} a_n (x - x_0)^n$ ...(4.4)
be the series solution of the given equation.

(ii) Differentiate y w.r.t. to x twice and substitute y, y', y'' in Eq. (4.3).

(iii) Shift the summation index to obtain a common power of x in each term.

(iv) Equate the coefficients of various powers of x to zero to obtain $a_1, a_2, a_3, \ldots$ in terms of a_0.

(v) Substitute $a_1, a_2, a_3, \ldots$ in Eq. (4.3) to obtain the required solution of the given equation.

Example 1

Find the series solution of $y' - 2xy = 0$. **[Winter 2014]**

Solution

$$y' - 2xy = 0 \qquad \ldots(1)$$

$$P(x) = 1, \qquad Q(x) = -2x$$

Since $P(x)$ and $Q(x)$ are both analytic at $x = 0$, it is an ordinary point.
Let the series solution of Eq. (1) about $x = 0$ be

$$y = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \cdots \qquad \ldots(2)$$

$$y' = \sum_{n=0}^{\infty} a_n n\, x^{n-1} = \sum_{n=1}^{\infty} n\, a_n x^{n-1}$$

Substituting in Eq. (1),

$$\sum_{n=1}^{\infty} n\, a_n x^{n-1} - 2x \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\sum_{n=1}^{\infty} n\, a_n x^{n-1} - 2 \sum_{n=0}^{\infty} a_n x^{n+1} = 0$$

To obtain common power of x in each term, putting $n - 1 = m + 1$ (i.e., $n = m + 2$) in the first term, we get

$$\sum_{m=-1}^{\infty} (m+2)\, a_{m+2} x^{m+1} - 2 \sum_{n=0}^{\infty} a_n x^{n+1} = 0$$

Since m is a dummy variable, replacing m by n,

$$\sum_{n=-1}^{\infty} (n+2) a_{n+2} x^{n+1} - 2 \sum_{n=0}^{\infty} a_n x^{n+1} = 0$$

$$a_1 + \sum_{n=0}^{\infty} (n+2) a_{n+2} x^{n+1} - 2 \sum_{n=0}^{\infty} a_n x^{n+1} = 0$$

Equating the constant term and the coefficient of x^{n+1} to zero,

$$a_1 = 0$$

and $$(n+2) a_{n+2} - 2a_n = 0, \qquad n \geq 0$$

$$a_{n+2} = \frac{2}{n+2} a_n, \qquad n \geq 0$$

Putting $n = 0, 1, 2, 3, \ldots$

$$a_2 = a_0$$

$$a_3 = \frac{2}{3}a_1 = 0 \qquad [\because\ a_1 = 0]$$

$$a_4 = \frac{2}{4}a_2 = \frac{1}{2}a_0$$

$$a_5 = \frac{2}{5}a_3 = 0$$

$$a_6 = \frac{2}{6}a_4 = \frac{2}{6}\cdot\frac{1}{2}a_0 = \frac{1}{6}a_0 = \frac{1}{3!}a_0$$

and so on.

Substituting in Eq. (2),

$$y = a_0 + 0.x + a_0x^2 + 0\cdot x^3 + \frac{1}{2}a_0x^4 + 0\cdot x^5 + \frac{1}{3!}a_0x^6 + \cdots$$

$$= a_0\left(1 + x^2 + \frac{x^4}{2!} + \frac{x^6}{3!} + \cdots\right)$$

Example 2

Find the power-series solution of the equation

$$\frac{d^2y}{dx^2} + y = 0 \quad \textit{about} \quad x_0 = 0. \qquad \textbf{[Summer 2014]}$$

Solution

$$y'' + y = 0 \qquad \ldots(1)$$

$P_0(x) = 1 \neq 0$ at $x = 0$

Hence, $x = 0$ is an ordinary point.

Let the series solution of Eq. (1) about $x = 0$ be

$$y = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1x + a_2x^2 + a_3x^3 + \cdots$$

$$y' = \sum_{n=0}^{\infty} a_n n x^{n-1} = \sum_{n=1}^{\infty} n a_n x^{n-1} \qquad \ldots(2)$$

$$y'' = \sum_{n=0}^{\infty} a_n n(n-1)x^{n-2} = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2}$$

Substituting in Eq. (1),

$$\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + \sum_{n=0}^{\infty} a_n x^n = 0$$

To obtain a common power of x in each term, putting $n-2=m$ (i.e., $n=m+2$) in the first term, we get

$$\sum_{m=0}^{\infty}(m+2)(m+1)a_{m+2}x^m+\sum_{n=0}^{\infty}a_nx^n=0$$

Since m is a dummy variable, replacing m by n in the first term,

$$\sum_{n=0}^{\infty}(n+2)(n+1)a_{n+2}x^n+\sum_{n=0}^{\infty}a_nx^n=0$$

Equating the coefficient of x^n to zero,

$$(n+1)(n+2)a_{n+2}+a_n=0, \qquad n\geq 0$$

$$a_{n+2}=-\frac{1}{(n+1)(n+2)}a_n, \qquad n\geq 0$$

Putting $n=0, 1, 2, \ldots$

$$a_2=-\frac{1}{1\cdot 2}a_0=-\frac{1}{2!}a_0$$

$$a_3=-\frac{1}{2\cdot 3}a_1=-\frac{1}{3!}a_1$$

$$a_4=-\frac{1}{3\cdot 4}a_2=-\frac{1}{3\cdot 4}\left(-\frac{1}{1\cdot 2}a_0\right)=\frac{1}{4!}a_0$$

$$a_5=-\frac{1}{4\cdot 5}a_3=-\frac{1}{4\cdot 5}\left(-\frac{1}{3!}a_1\right)=\frac{1}{5!}a_1$$

and so on.

Substituting in Eq. (2),

$$y=a_0+a_1x-\frac{1}{2!}a_0x^2-\frac{1}{3!}a_1x^3+\frac{1}{4!}a_0x^4+\frac{1}{5!}a_1x^5+\cdots$$

$$=a_0\left(1-\frac{x^2}{2!}+\frac{x^4}{4!}-\cdots\right)+a_1\left(x-\frac{x^3}{3!}+\frac{x^5}{5!}-\cdots\right)$$

$$=a_0\cos x+a_1\sin x$$

Example 3

Find the series solution of $y''=2y'$ in powers of x. **[Winter 2012]**

Solution

$$y''=2y'$$

$$y''-2y'=0 \qquad \ldots(1)$$

$$P_0(x)=1\neq 0 \text{ at } x=0$$

Hence, $x=0$ is an ordinary point.

Let the series solution of Eq. (1) in powers of x be

$$y = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \cdots \qquad \ldots(2)$$

$$y' = \sum_{n=0}^{\infty} a_n n x^{n-1} = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

$$y'' = \sum_{n=0}^{\infty} a_n n(n-1) x^{n-2} = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

Substituting in Eq. (1),

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} - 2\sum_{n=1}^{\infty} n a_n x^{n-1} = 0$$

To obtain a common power of x in each term, putting $n - 2 = m - 1$ (i.e., $n = m + 1$) in the first term, we get

$$\sum_{m=1}^{\infty} (m+1)\, m\, a_{m+1} x^{m-1} - 2\sum_{n=1}^{\infty} n a_n x^{n-1} = 0$$

Since m is a dummy variable, replacing m by n,

$$\sum_{n=1}^{\infty} n(n+1) a_{n+1} x^{n-1} - 2\sum_{n=1}^{\infty} n a_n x^{n-1} = 0$$

Equating the coefficient of x^{n-1} to zero,

$$n(n+1)a_{n+1} - 2n\, a_n = 0, \qquad n \geq 1$$

$$a_{n+1} = -\frac{2n}{n(n+1)} a_n, \qquad n \geq 1$$

$$= \frac{2}{n+1} a_n, \qquad n \geq 1$$

Putting $n = 1, 2, 3, \ldots$

$$a_2 = \frac{2}{2} a_1 = a_1$$

$$a_3 = \frac{2}{3} a_2 = \frac{2}{3} a_1$$

$$a_4 = \frac{2}{4} a_3 = \frac{1}{2} \cdot \frac{2}{3} a_1 = \frac{1}{3} a_1$$

and so on.

Substituting in Eq. (2),

$$y = a_0 + a_1 x + a_1 x^2 + \frac{2}{3} a_1 x^3 + \frac{1}{3} a_1 x^4 + \cdots$$

$$= a_0 + a_1 \left(x + x^2 + \frac{2}{3} x^3 + \frac{1}{3} x^4 + \cdots \right)$$

Example 4

Find the power-series solution of $\dfrac{d^2y}{dx^2} + xy = 0$.

Solution

$$y'' + xy = 0 \qquad \ldots(1)$$

$P_0(x) = 1 \neq 0$ at $x = 0$

Hence, $x = 0$ is an ordinary point about $x = 0$.

Let the series solution of Eq. (1) be

$$y = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \cdots \qquad \ldots(2)$$

$$y' = \sum_{n=0}^{\infty} a_n \cdot n x^{n-1} = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

$$y'' = \sum_{n=0}^{\infty} a_n n(n-1) x^{n-2} = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

Substituting in Eq. (1),

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} + x \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} + \sum_{n=0}^{\infty} a_n x^{n+1} = 0$$

To obtain the common power of x in each term, putting $n - 2 = m + 1$ (i.e., $n = m + 3$) in the first term, we get

$$\sum_{m=-1}^{\infty} (m+3)(m+2) a_{m+3} x^{m+1} + \sum_{n=0}^{\infty} a_n x^{n+1} = 0$$

Since m is a dummy variable, replacing m by n,

$$\sum_{n=-1}^{\infty} (n+3)(n+2) a_{n+3} x^{n+1} + \sum_{n=0}^{\infty} a_n x^{n+1} = 0$$

$$2a_2 + \sum_{n=0}^{\infty} (n+2)(n+3) a_{n+3} x^{n+1} + \sum_{n=0}^{\infty} a_n x^{n+1} = 0$$

Equating the constant term and the coefficient of x^{n+1} to zero,

$$2a_2 = 0, \qquad a_2 = 0$$

and $$(n+2)(n+3)\, a_{n+3} + a_n = 0, \qquad n \geq 0$$

$$a_{n+3} = -\frac{1}{(n+2)(n+3)} a_n, \qquad n \geq 0$$

Putting $n = 0, 1, 2, 3, 4, \ldots$

$$a_3 = -\frac{1}{2\cdot 3}a_0$$

$$a_4 = -\frac{1}{3\cdot 4}a_1$$

$$a_5 = -\frac{1}{4\cdot 5}a_2 = 0 \qquad [\because\ a_2 = 0]$$

$$a_6 = -\frac{1}{5\cdot 6}a_3 = -\frac{1}{5\cdot 6}\left(-\frac{1}{2\cdot 3}\right)a_0 = \frac{1}{2\cdot 3\cdot 5\cdot 6}a_0$$

$$a_7 = -\frac{1}{6\cdot 7}a_4 = -\frac{1}{6\cdot 7}\left(-\frac{1}{3\cdot 4}a_1\right) = \frac{1}{3\cdot 4\cdot 6\cdot 7}a_1$$

and so on.

Substituting in Eq. (2),

$$y = a_0 + a_1 x + 0\cdot x^2 - \frac{1}{2\cdot 3}a_0 x^3 - \frac{1}{3\cdot 4}a_1 x^4 + 0\cdot x^5 + \frac{1}{2\cdot 3\cdot 5\cdot 6}a_0 x^6 + \frac{1}{3\cdot 4\cdot 6\cdot 7}a_1 x^7 + \cdots$$

$$= a_0\left(1 - \frac{x^3}{2\cdot 3} + \frac{x^6}{2\cdot 3\cdot 5\cdot 6} - \cdots\right) + a_1\left(x - \frac{x^4}{3\cdot 4} + \frac{x^7}{3\cdot 4\cdot 6\cdot 7}\right)\cdots$$

Example 5

Solve in series the equation $\dfrac{d^2y}{dx^2} + x^2 y = 0$. **[Winter 2013]**

Solution

$$y'' + x^2 y = 0 \qquad \ldots(1)$$

$P_0(x) = 1 \neq 0$ at $x = 0$

Hence, $x = 0$ is an ordinary point.

Let the series solution of Eq. (1) about $x = 0$ be

$$y = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \cdots \qquad \ldots(2)$$

$$y' = \sum_{n=0}^{\infty} a_n n x^{n-1} = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

$$y'' = \sum_{n=0}^{\infty} a_n\, n(n-1)x^{n-2} = \sum_{n=2}^{\infty} n(n-1)a_n x^{n-2}$$

Substituting in Eq. (1),

$$\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + x^2 \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + \sum_{n=0}^{\infty} a_n x^{n+2} = 0$$

To obtain a common power of x in each term, putting $n - 2 = m + 2$ (i.e., $n = m + 4$) in the first term, we get

$$\sum_{m=-2}^{\infty} (m+4)(m+3)a_{m+4} x^{m+2} + \sum_{n=0}^{\infty} a_n x^{n+2} = 0$$

Since m is a dummy variable, replacing m by n,

$$\sum_{n=-2}^{\infty} (n+3)(n+4)a_{n+4} x^{n+2} + \sum_{n=0}^{\infty} a_n x^{n+2} = 0$$

$$2a_2 + 6a_3 x + \sum_{n=0}^{\infty} (n+3)(n+4)a_{n+4} x^{n+2} + \sum_{n=0}^{\infty} a_n x^{n+2} = 0$$

Equating the constant term, the coefficient of x, and the coefficient of x^{n+2} to zero,

$$2a_0 = 0, \qquad a_2 = 0$$

$$6a_3 = 0, \qquad a_3 = 0$$

and $$(n + 3)(n + 4)a_{n+4} + a_n = 0, \quad n \geq 0$$

$$a_{n+4} = -\frac{1}{(n+3)(n+4)} a_n, \qquad n \geq 0$$

Putting $n = 0, 1, 2, \ldots$

$$a_4 = -\frac{1}{3 \cdot 4} a_0$$

$$a_5 = -\frac{1}{4 \cdot 5} a_1$$

$$a_6 = -\frac{1}{5 \cdot 6} a_2 = 0 \quad [\because\ a_2 = 0]$$

$$a_7 = -\frac{1}{6 \cdot 7} a_3 = 0 \quad [\because\ a_3 = 0]$$

$$a_8 = -\frac{1}{7 \cdot 8} a_4 = -\frac{1}{7 \cdot 8}\left(-\frac{1}{3 \cdot 4} a_0\right) = \frac{1}{3 \cdot 4 \cdot 7 \cdot 8} a_0$$

$$a_9 = -\frac{1}{8 \cdot 9} a_5 = -\frac{1}{8 \cdot 9}\left(-\frac{1}{4 \cdot 5} a_1\right) = \frac{1}{4 \cdot 5 \cdot 8 \cdot 9} a_1$$

and so on.

Substituting in Eq. (2),

$$y = a_0 + a_1 x + 0 \cdot x^2 + 0 \cdot x^3 - \frac{1}{3 \cdot 4} a_0 x^4 - \frac{1}{4 \cdot 5} a_1 x^5 + 0 \cdot x^6 + 0 \cdot x^7 + \frac{1}{3 \cdot 4 \cdot 7 \cdot 8} a_0 x^8 + \frac{1}{4 \cdot 5 \cdot 8 \cdot 9} a_1 x^9 + \cdots$$

$$= a_0 \left(1 - \frac{x^4}{3 \cdot 4} + \frac{x^8}{3 \cdot 4 \cdot 7 \cdot 8} - \cdots\right) + a_1 \left(x - \frac{x^5}{4 \cdot 5} + \frac{x^9}{4 \cdot 5 \cdot 8 \cdot 9} - \cdots\right)$$

Example 6

Using the power-series method, solve

$$(1 - x^2)y'' - 2xy' + 2y = 0 \qquad \textbf{[Winter 2013]}$$

Solution

$$(1 - x^2)y'' - 2xy' + 2y = 0 \qquad \ldots(1)$$

$$P_0(x) = 1 - x^2 \neq 0 \quad \text{at } x = 0$$

Hence, $x = 0$ is an ordinary point.

Let the series solution of Eq. (1) about $x = 0$ be

$$y = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \cdots \qquad \ldots(2)$$

$$y' = \sum_{n=0}^{\infty} a_n n x^{n-1} = \sum_{n=1}^{\infty} n a_n x^{n-1}$$

$$y'' = \sum_{n=0}^{\infty} a_n \, n(n-1) x^{n-2} = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

Substituting in Eq. (1),

$$(1 - x^2) \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} - 2x \sum_{n=1}^{\infty} n a_n x^{n-1} + 2 \sum_{n=0}^{\infty} a_n x^n = 0$$

$$\sum_{n=2}^{\infty} n(n-1) a_n x^{n-2} - \sum_{n=2}^{\infty} n(n-1) a_n x^n - 2 \sum_{n=1}^{\infty} n \, a_n x^n + 2 \sum_{n=0}^{\infty} a_n x^n = 0$$

To obtain a common power of x in each term, putting $n - 2 = m$ in the first term, we get

$$\sum_{n=0}^{\infty} (m+2)(m+1) a_{m+2} x^m - \sum_{n=2}^{\infty} n(n-1) a_n x^n - 2 \sum_{n=1}^{\infty} n a_n x^n + 2 \sum_{n=0}^{\infty} a_n x^n = 0$$

Since m is a dummy variable, replacing m by n,

$$\sum_{n=0}^{\infty}(n+2)(n+1)a_{n+2}x^n - \sum_{n=2}^{\infty}n(n-1)a_nx^n - 2\sum_{n=1}^{\infty}na_nx^n + 2\sum_{n=0}^{\infty}a_nx^n = 0$$

$$2a_2 + 6a_3x + \sum_{n=2}^{\infty}(n+2)(n+1)a_{n+2}x^n - \sum_{n=2}^{\infty}n(n-1)a_nx^n - 2a_1x - 2\sum_{n=2}^{\infty}na_nx^n$$

$$+ 2a_0 + 2a_1x + 2\sum_{n=2}^{\infty}a_nx^n = 0$$

$$2a_0 + 2a_2 + 6a_3x + \sum_{n=2}^{\infty}\left[(n+1)(n+2)a_n + 2 - n(n-1)a_n - 2na_n + 2a_n\right]x^n = 0$$

$$2(a_0 + a_2) + 6a_3x + \sum_{n=2}^{\infty}\left[(n+1)(n+2)a_{n+2} - (n-1)(n+2)a_n\right]x^n = 0$$

Equating the constant term, the coefficient of x, and the coefficient of x^n to zero,

$$2a_2 + 2a_0 = 0$$

$$a_2 = -a_0$$

$$3 \cdot 2a_3x = 0$$

$$a_3 = 0$$

and $$(n+2)(n+1)a_{n+2} - n(n-1)a_n - 2na_n + 2a_n = 0, \quad n \geq 0$$

$$(n+1)(n+2)a_{n+2} - (n^2 + n - 2)a_n = 0, \quad n \geq 0$$

$$a_{n+2} = \frac{(n-1)(n+2)}{(n+1)(n+2)}a_n, \quad n \geq 0$$

$$= \frac{n-1}{n+1}a_n, \quad n \geq 0$$

Putting $n = 2, 3, \ldots$

$$a_4 = \frac{1}{3}a_2 = -\frac{1}{3}a_0$$

$$a_5 = \frac{2}{4}a_3 = 0$$

$$a_6 = \frac{3}{5}a_4 = -\frac{1}{5}a_0$$

and so on.

Substituting these values in Eq. (2),

$$y = a_0 + a_1x - a_0x^2 + 0 \cdot x^3 - \frac{1}{3}a_0x^4 + 0 \cdot x^5 + \frac{1}{5}a_0x^6 + \cdots$$

$$y = a_0 + a_1x - a_0x^2 - \frac{1}{3}a_0x^4 - \frac{1}{5}a_0x^5 \cdots$$

$$= a_1 x + a_0\left(1 - x^2 - \frac{1}{3}x^4 - \frac{1}{5}x^6 \cdots\right)$$

where a_0 and a_1 are arbitrary constants.

Example 7

Find the power-series solution of the equation

$(x^2 + 1)\,y'' + xy' - xy = 0$ about $x = 0$.

[Winter 2012, Summer 2013]

Solution

$$(x^2 + 1)\,y'' + xy' - xy = 0 \qquad \ldots(1)$$

$P_0(x) = 1 + x^2 = 1 \neq 0$ at $x = 0$

At $x = 0$, $P_0(0) = 1 \neq 0$

Hence, $x = 0$ is an ordinary point.

Let the series solution of Eq. (1) about $x = 0$ be

$$y = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \cdots \qquad \ldots(2)$$

$$y' = \sum_{n=0}^{\infty} a_n n\, x^{n-1} = \sum_{n=1}^{\infty} n\, a_n x^{n-1}$$

$$y'' = \sum_{n=0}^{\infty} a_n n(n-1) x^{n-2} = \sum_{n=2}^{\infty} n(n-1) a_n x^{n-2}$$

Substituting in Eq. (1),

$$(x^2+1)\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + x\sum_{n=1}^{\infty} n\, a_n x^{n-1} - x\sum_{n=0}^{\infty} a_n x^n = 0$$

$$\sum_{n=2}^{\infty} n(n-1)a_n (x^n + x^{n-2}) + \sum_{n=1}^{\infty} n\, a_n x^n - \sum_{n=0}^{\infty} a_n x^{n+1} = 0$$

$$\sum_{n=2}^{\infty} n(n-1)a_n x^{n-2} + \sum_{n=2}^{\infty} n(n-1)a_n x^n + \sum_{n=1}^{\infty} n\, a_n x^n - \sum_{n=0}^{\infty} a_n x^{n+1} = 0$$

To obtain a common power of x in each term, putting $n - 2 = m$ in the first term and $n + 1 = t$ in the fourth term, we get

$$\sum_{m=0}^{\infty} (m+2)(m+1)a_{m+2} x^m + \sum_{n=2}^{\infty} n(n-1)a_n x^n + \sum_{n=1}^{\infty} n\, a_n x^n - \sum_{t=1}^{\infty} a_{t-1} x^t = 0$$

Since m and t are dummy variables, replacing m and t by n,

$$\sum_{n=0}^{\infty}(n+1)(n+2)a_{n+2}x^n+\sum_{n=2}^{\infty}n(n-1)a_nx^n+\sum_{n=1}^{\infty}n\,a_nx^n-\sum_{n=1}^{\infty}a_{n-1}x^n=0$$

$$\left[2a_2+6a_3x+\sum_{n=2}^{\infty}(n+1)(n+2)a_{n+2}x^n\right]+\sum_{n=2}^{\infty}n(n-1)a_nx^n$$

$$+\left[a_1x+\sum_{n=2}^{\infty}n\,a_nx^n\right]-\left[a_0x+\sum_{n=2}^{\infty}a_{n-1}x^n\right]=0$$

$$2a_2+(6a_3+a_1-a_0)x+\sum_{n=2}^{\infty}\left[(n+1)(n+2)a_{n+2}+n(n-1)a_n+na_n-a_{n-1}\right]x^n=0$$

$$2a_2+(6a_3+a_1-a_0)x+\sum_{n=2}^{\infty}\left[(n+1)(n+2)a_{n+2}+n^2a_n-a_{n-1}\right]x^n=1$$

Equating the constant term, the coefficients of x and x^n to zero,

$$2a_2=0,\qquad a_2=0$$

$$6a_3+a_1-a_0=0,\qquad a_3=\frac{1}{6}(a_0-a_1)$$

and $(n+1)(n+2)\,a_{n+2}+n^2a_n-a_{n-1}=0,\ n\geq 2$

$$a_{n+2}=\frac{a_{n-1}-n^2a_n}{(n+1)(n+2)},\qquad n\geq 2$$

Putting $n = 2, 3, 4, \ldots$

$$a_4=\frac{a_1-4a_2}{12}=\frac{a_1}{12}$$

$$a_5=\frac{a_2-9a_3}{20}=-\frac{9}{20}\left[\frac{1}{6}(a_0-a_1)\right]=\frac{3}{40}(a_1-a_0)$$

and so on.

Substituting in Eq. (2),

$$y=a_0+a_1x+0\cdot x^2+\frac{1}{6}(a_0-a_1)x^3+\frac{a_1}{12}x^4+\frac{3}{40}(a_1-a_0)x^5+\cdots$$

$$=a_0\left(1+\frac{x^3}{6}-\frac{3}{40}x^5+\cdots\right)+a_1\left(x-\frac{x^3}{6}+\frac{x^4}{12}+\frac{3}{40}x^5+\cdots\right)$$

Example 8

Solve the initial-value problem

$$x\frac{d^2y}{dx^2}+\frac{dy}{dx}+2y=0\qquad \textit{with}\quad y(1)=2,\ y'(1)=4$$

Solution

$$xy'' + y' + 2y = 0 \qquad \ldots(1)$$

Since the initial conditions are given at $x = 1$, a power-series solution of Eq. (1) in powers of $(x - 1)$ is obtained.

$$P_0(x) = x \neq 0 \text{ at } x = 1$$

Let the series solution of Eq. (1) be

$$y = \sum_{n=0}^{\infty} a_n (x-1)^n = a_0 + a_1(x-1) + a_2(x-1)^2 + \cdots \qquad \ldots(2)$$

Let $x - 1 = t$

$$y' = \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{dy}{dt} \cdot 1 = \frac{dy}{dt}$$

$$y'' = \frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dt}\right) = \frac{d}{dx}\left(\frac{dy}{dt}\right) = \frac{d}{dt}\left(\frac{dy}{dt}\right) \cdot \frac{dt}{dx} = \frac{d^2y}{dt^2}$$

Substituting in Eq. (1),

$$(t+1)\frac{d^2y}{dt^2} + \frac{dy}{dt} + 2y = 0 \qquad \ldots(3)$$

Putting $x - 1 = t$ in Eq. (2), the series solution of Eq. (3) is

$$y = \sum_{n=0}^{\infty} a_n n = a_0 + a_1 t + a_2 t^2 + \cdots \qquad \ldots(4)$$

$$\frac{dy}{dt} = \sum_{n=0}^{\infty} a_n \cdot n t^{n-1} = \sum_{n=1}^{\infty} n a_n t^{n-1}$$

$$\frac{d^2y}{dt^2} = \sum_{n=0}^{\infty} a_n \cdot n(n-1) t^{n-2} = \sum_{n=2}^{\infty} n(n-1) a_n t^{n-2}$$

Substituting in Eq. (3),

$$(1+t)\sum_{n=2}^{\infty} n(n-1)a_n t^{n-2} + \sum_{n=1}^{\infty} n a_n t^{n-1} + 2\sum_{n=0}^{\infty} a_n t^n = 0$$

$$\sum_{n=2}^{\infty} n(n-1)a_n t^{n-2} + \sum_{n=2}^{\infty} n(n-1)a_n t^{n-1} + \sum_{n=1}^{\infty} n\, a_n t^{n-1} + 2\sum_{n=0}^{\infty} a_n t^n = 0$$

To obtain a common power of t, putting $n - 2 = m$, in the first term and $n - 1 = m_2$ in the second and third terms, we get

$$\sum_{m_1=0}^{\infty} (m_1+2)(m_1+1)a_{m_1+2} t^{m_1} + \sum_{m_2=1}^{\infty} (m_2+1)m_2 a_{m_2+1} t^{m_2}$$

$$+ \sum_{m_2=0}^{\infty} (m_2+1)a_{m_2+1} t^{m_2} + 2\sum_{n=0}^{\infty} a_n t^n = 0$$

Since m_1 and m_2 are dummy variables, replacing m_1 and m_2 by n,

$$\sum_{n=0}^{\infty}(n+1)(n+2)a_{n+2}t^n+\sum_{n=1}^{\infty}n(n+1)a_{n+1}t^n+\sum_{n=0}^{\infty}(n+1)a_{n+1}t^n+2\sum_{n=0}^{\infty}a_nt^n=0$$

$$2a_2+\sum_{n=1}^{\infty}(n+1)(n+2)a_{n+2}t^n+\sum_{n=1}^{\infty}n(n+1)a_{n+1}t^n+a_1$$

$$+\sum_{n=1}^{\infty}(n+1)a_{n+1}t^n+2a_0+2\sum_{n=1}^{\infty}a_nt^n=0$$

$$2a_2+a_1+2a_0+\sum_{n=1}^{\infty}\left[(n+1)(n+2)a_{n+2}+(n+1)^2a_{n+1}+2a_n\right]t^n=0$$

Equating the constant term and the coefficient of t^n to zero,

$$2a_2+a_1+2a_0=0$$

$$a_2=-\frac{1}{2}(a_1+2a_0)$$

and $$(n+1)(n+2)a_{n+2}+(n+1)^2a_{n+1}+2a_n=0,\quad n\geq 1$$

$$a_{n+2}=-\frac{(n+1)^2a_{n+1}+2a_n}{(n+1)(n+2)},\quad n\geq 1$$

Putting $n = 1, 2, 3, \ldots$

$$a_3=-\frac{4a_2+2a_1}{2\cdot 3}=\frac{1}{6}\left[-2(a_1+2a_0)+2a_1\right]=\frac{2}{3}a_0$$

$$a_4=-\frac{9a_3+2a_2}{3\cdot 4}=-\frac{1}{12}\left[9\cdot\frac{2}{3}a_0-(a_1+2a_0)\right]=-\frac{1}{12}(4a_0-a_1)$$

and so on.

Substituting in Eq. (4),

$$y=a_0+a_1t-\left(\frac{a_1+2a_0}{2}\right)t^2+\frac{2}{3}a_0t^3-\frac{1}{12}(4a_0-a_1)t^4+\cdots$$

$$=a_0+a_1(x-1)-\frac{1}{2}(a_1+2a_0)(x-1)^2+\frac{2}{3}a_0(x-1)^3-\frac{1}{12}(4a_0-a_1)(x-1)^4+\cdots \quad \ldots(5)$$

$$y'=a_1-(a_1+2a_0)(x-1)+2a_0(x-1)^2-\frac{1}{3}(4a_0-a_1)(x-1)^3+\cdots \quad \ldots(6)$$

Initially, at $x = 1$, $y = 2$, and $y' = 4$

Substituting in Eqs (5) and (6),

$$2=a_0$$

and $$4=a_1$$

Putting in Eq. (5),

$$y=2+4(x-1)-4(x-1)^2+\frac{4}{3}(x-1)^3-\frac{1}{3}(x-1)^4+\cdots$$

10. $y'' - xy' + (2x^2 + 1)y = 0$

$$\left[\text{Ans.}: y = a_0\left(1 - \frac{x^2}{2} - \frac{5}{24}x^4 + \cdots\right) + a_1\left(x - \frac{x^5}{10} + \cdots\right)\right]$$

Find the power-series solutions of the following equations about the given point.

11. $xy' - y = 0, \quad x_0 = 1$

$$\left[\text{Ans.}: y = a_0[1 + (x-1)]\right]$$

12. $y'' + xy' + y = 0, \quad x_0 = 2$

$$\left[\text{Ans.}: y = a_0\left[1 - \frac{1}{2}(x-2)^2 + \frac{1}{3}(x-2)^3 + \cdots\right] + a_1\left[(x-2) - (x-2)^2 + \frac{1}{3}(x-2)^2 \cdots\right]\right]$$

13. $(x+1)y' - (x+2)y = 0, \quad x_0 = -2$

$$\left[\text{Ans.}: y = a_0\left[1 - \frac{1}{2}(x+2)^2 - \frac{1}{3}(x-2)^3 - \cdots\right]\right]$$

Find the power-series solutions of the following initial-value equations.

14. $y'' - xy = 0, y(1) = 2, y'(1) = 0$

$$\left[\text{Ans.}: y = 2 + (x-1)^2 + \frac{1}{3}(x-1)^3 + \cdots\right]$$

15. $(x^2 + 2)y'' - 2xy' + 3y = 0, y(1) = 1, y'(1) = -1$

$$\left[\text{Ans.}: y = 1 - (x-1) - \frac{5}{6}(x-1)^2 + \cdots\right]$$

4.4 FROBENIUS METHOD

In the previous section, the power-series solution for differential equations is obtained when x_0 is an ordinary point. To obtain the solution near a regular singular point x_0, an extension of the power-series method, known as the Frobenius method (or generalised power-series method), is used.

Let x_0 be a regular singular point of the differential equation

$$P_0(x)y'' + P_1(x)y' + P_2(x)y = 0 \qquad \ldots(4.5)$$

$$y'' + P(x)y' + Q(x)y = 0$$

where $P(x) = \dfrac{P_1(x)}{P_0(x)}, \quad Q(x) = \dfrac{P_2(x)}{P_0(x)}$

(i) Let the series solution of Eq. (4.5) about x_0 be

$$y = \sum_{n=0}^{\infty} a_n (x - x_0)^{n+r} = (x - x_0)^r \left[a_0 + a_1 (x - x_0) + a_2 (x - x_0)^2 + \cdots \right] \quad \ldots(4.6)$$

(ii) Differentiate twice and substitute y, y' and y'' in Eq. (4.5).

(iii) Equating to zero the coefficients of the lowest degree term in $(x - x_0)$, a quadratic equation, known as *indicial equation*, is obtained. The roots of the indicial equation are called *indicial roots*.

(iv) Equating to zero the coefficients of other powers, of x, a recurrence relation relating the coefficients a_n is obtained.

(v) Using the recurrence relation for each indicial root separately, two linearly independent solutions $y_1(x)$ and $y_2(x)$ of Eq. (4.5) are obtained.
The general solution of Eq. (4.5) is given as

$$y(x) = c_1 y_1(x) + c_2 y_2(x)$$

where c_1 and c_2 are arbitrary constants.

(vi) One of the solutions $y_1(x)$ or $y_2(x)$ is in the form of Eq. (4.6). The form of the other solution depends upon the nature of the indicial roots.

Let r_1 and r_2 be the roots of the indicial equation. There are three cases.

Case I Distinct Roots not Differing by an Integer

$$r_1 - r_2 \neq \text{an integer}$$

Then $y_1 = (y)_{r=r_1}$ and $y_2 = (y)_{r=r_2}$

The general solution is

$$y = c_1 y_1 + c_2 y_2$$

Case II Double Root (Repeated Root)

$$r_1 = r_2 = b, \text{ say}$$

$$y_1 = (y)_{r=b} \quad \text{and} \quad y_2 = \left(\frac{\partial y}{\partial r} \right)_{r=b}$$

The general solution is

$$y = c_1 y_1 + c_2 y_2$$

$$= c_1 (y)_{r=b} + c_2 \left(\frac{\partial y}{\partial r} \right)_{r=b}$$

Case III Roots Differing by an Integer

$$r_1 - r_2 = \text{an integer}, \quad r_1 < r_2$$

In this case, solutions corresponding to r_1 and r_2 may or may not be linearly independent. This leads to two possibilities:

(i) One of the coefficients becomes infinite for the smaller indicial root $r = r_1$.

The procedure is modified by putting $a_0 = c_0\,(r - r_1),\ c_0 \neq 0$

$$y_1(y)_{r=r_1} \quad \text{and} \quad y_2 = \left(\frac{\partial y}{\partial r}\right)_{r=r_1}$$

The solution, corresponding to the second indicial root r_2 is usually a multiple of y_1 or a part of $\left(\frac{\partial y}{\partial r}\right)_{r=r_1}$. Hence, it produces a linearly dependent solution.

The general solution is

$$y = c_1 y_1 + c_2 y_2$$

$$= c_1 (y)_{r=r_1} + c_2 \left(\frac{\partial y}{\partial r}\right)_{r=r_1}$$

(ii) One of the coefficients becomes indeterminate for the smaller indicial root $r = r_1$. This root produces the complete solution as it contains two arbitrary constants. The second indicial root r_2 produces a linearly dependent solution.

Example 1

Solve in series the differential equation

$$4x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0$$

[Winter 2014]

Solution

$$4xy'' + 2y' + y = 0 \qquad \ldots(1)$$

$P_0(x) = 4x = 0$ at $x = 0$

Hence, $x = 0$ is a singular point.

$$y'' + \frac{1}{2x}y' + \frac{1}{4x}y = 0$$

$$P(x) = \frac{1}{2x}, \qquad Q(x) = \frac{1}{4x}$$

Since $xP(x) = \frac{1}{2}$ and $x^2Q(x) = \frac{x}{4}$ are analytic (i.e., differentiable) at $x = 0$, it is a regular singular point.

Let the series solution of Eq. (1) about $x = 0$ be

$$y = \sum_{n=0}^{\infty} a_n x^{n+r} = x^r (a_0 + a_1 x + a_2 x^2 + \cdots) \qquad \ldots(2)$$

$$y' = \sum_{n=0}^{\infty} a_n (n+r) x^{n+r-1}$$

$$y'' = \sum_{n=0}^{\infty} a_n (n+r)(n+r-1) x^{n+r-2}$$

Substituting in Eq. (1),

$$4x\sum_{n=0}^{\infty}(n+r)(n+r-1)a_n x^{n+r-2}+2\sum_{n=0}^{\infty}(n+r)a_n x^{n+r-1}+\sum_{n=0}^{\infty}a_n x^{n+r}=0$$

$$4\sum_{n=0}^{\infty}(n+r)(n+r-1)a_n x^{n+r-1}+2\sum_{n=0}^{\infty}(n+r)a_n x^{n+r-1}+\sum_{n=0}^{\infty}a_n x^{n+r}=0$$

$$\sum_{n=0}^{\infty}2(n+r)(2n+2r-1)a_n x^{n+r-1}+\sum_{n=0}^{\infty}a_n x^{n+r}=0$$

To obtain a common power of x in each term, putting $n = m + 1$ in the first term, we get

$$\sum_{m=-1}^{\infty}2(m+1+r)(2m+2+2r-1)a_{m+1}x^{m+r}+\sum_{n=0}^{\infty}a_n x^{n+r}=0$$

Since m is a dummy variable, replacing m by n

$$\sum_{n=-1}^{\infty}2(n+r+1)(2n+2r+1)a_{n+1}x^{n+r}+\sum_{n=0}^{\infty}a_n x^{n+r}=0$$

$$2r(-2+2r+1)a_0x^{-1+r}+\sum_{n=0}^{\infty}2(n+r+1)(2n+2r+1)a_{n+1}x^{n+r}+\sum_{n=0}^{\infty}a_n x^{n+r}=0$$

$$2r(2r-1)a_0x^{r-1}+\sum_{n=0}^{\infty}\left[2(n+r+1)(2n+2r+1)a_{n+1}+a_n\right]x^{n+r}=0$$

Equating the coefficient of the lowest degree term to zero, the indicial equation is

$$2a_0r(2r-1)=0$$

$$r=0,\qquad r=\frac{1}{2}\quad[\because\ a_0\neq 0]$$

Equating the coefficient of x^{n+r} to zero,

$$2(n+r+1)(2n+2r+1)a_{n+1}+a_n=0,\qquad n\geq 0$$

$$a_{n+1}=-\frac{1}{2(n+r+1)(2n+2r+1)}a_n,\qquad n\geq 0 \qquad ...(3)$$

For the first solution, putting $r = 0$ in Eq. (3),

$$a_{n+1}=-\frac{1}{2(n+1)(2n+1)}a_n,\qquad n\geq 0$$

Putting $n = 0, 1, 2, ...$

$$a_1=-\frac{1}{2}a_0$$

$$a_2=-\frac{1}{2\cdot 2\cdot 3}a_1=-\frac{1}{12}\left(-\frac{1}{2}a_0\right)=\frac{1}{24}a_0$$

and so on.

Substituting $r = 0$ and values of $a_1, a_2, \ldots$ in Eq. (2),

$$y_1 = a_0 - \frac{1}{2}a_0 x + \frac{1}{24}a_0 x^2 - \cdots$$

$$= a_0\left(1 - \frac{x}{2} + \frac{x^2}{24} - \cdots\right)$$

For the second solution, putting $r = \frac{1}{2}$ in Eq. (3),

$$a_{n+1} = -\frac{1}{2\left(n + \frac{1}{2} + 1\right)(2n+1+1)}a_n, \qquad n \geq 0$$

$$= -\frac{1}{2\left(n + \frac{3}{2}\right)(2n+2)}a_n, \qquad n \geq 0$$

Putting $n = 0, 1, 2, \ldots$

$$a_1 = -\frac{1}{2 \cdot \frac{3}{2} \cdot 2}a_0 = -\frac{1}{6}a_0$$

$$a_2 = -\frac{1}{2 \cdot \frac{5}{2} \cdot 4}a_1 = -\frac{1}{20}\left(-\frac{1}{6}a_0\right) = \frac{1}{120}a_0$$

and so on.

Substituting $r = \frac{1}{2}$ and values of $a_1, a_2, \ldots$ in Eq. (2),

$$y_2 = x^{\frac{1}{2}}\left(a_0 - \frac{1}{6}a_0 x + \frac{1}{120}a_0 x^2 - \cdots\right)$$

$$= a_0\sqrt{x}\left(1 - \frac{x}{6} + \frac{x^2}{120} - \cdots\right)$$

Hence, the general solution is

$$y = c_1 y_1 + c_2 y_2$$

$$= c_1 a_0\left(1 - \frac{x}{2} + \frac{x^2}{24} + - \cdots\right) + c_2 a_0\sqrt{x}\left(1 - \frac{x}{6} + \frac{x^2}{120} - \cdots\right)$$

$$= A\left(1 - \frac{x}{2} + \frac{x^2}{24} - \cdots\right) + B\sqrt{x}\left(1 - \frac{x}{6} + \frac{x^2}{120} - \cdots\right)$$

$$= A\cos\sqrt{x} + B\sin\sqrt{x}$$

where $A = c_1 a_0, \quad B = c_2 a_0$

Example 2

Using the Forbenius method, obtain the series solution for

$$2x(1-x)y'' + (1-x)y' + 3y = 0 \quad \text{about} \quad x_0 = 0$$

Solution

$$2x(1-x)y'' + (1-x)y' + 3y = 0 \qquad \ldots(1)$$

$$P_0(x) = 2x\,(1-x) = 0 \text{ at } x = 0$$

Hence, $x = 0$ is a singular point.

$$y'' + \frac{1}{2x}y' + \frac{3}{2x(1-x)}y = 0$$

$$P(x) = \frac{1}{2x}, \qquad Q(x) = \frac{3}{2x(1-x)}$$

Since $xP(x) = \frac{1}{2}$ and $x^2Q(x) = \frac{3x}{2(1-x)}$ are analytic (i.e., differentiable) at $x = 0$, it is a regular singular point.

Let the series solution of Eq. (1) about $x = 0$ be

$$y = \sum_{n=0}^{\infty} a_n x^{n+r} = x^r(a_0 + a_1x + a_2x^2 + \cdots) \qquad \ldots(2)$$

$$y' = \sum_{n=0}^{\infty} a_n(n+r)x^{n+r-1}$$

$$y'' = \sum_{n=0}^{\infty} a_n(n+r)(n+r-1)x^{n+r-2}$$

Substituting in Eq. (1),

$$2x(1-x)\sum_{n=0}^{\infty}(n+r)(n+r-1)a_nx^{n+r-2} + (1-x)\sum_{n=0}^{\infty}(n+r)a_nx^{n+r-1} + 3\sum_{n=0}^{\infty}a_nx^{n+r} = 0$$

$$2\sum_{n=0}^{\infty}(n+r)(n+r-1)a_nx^{n+r-1} - 2\sum_{n=0}^{\infty}(n+r)(n+r-1)a_nx^{n+r} + \sum_{n=0}^{\infty}(n+r)a_nx^{n+r-1}$$

$$- \sum_{n=0}^{\infty}(n+r)a_nx^{n+r} + 3\sum_{n=0}^{\infty}a_nx^{n+r} = 0$$

$$\sum_{n=0}^{\infty}(n+r)(2n+2r-2+1)a_nx^{n+r-1} - \sum_{n=0}^{\infty}\left[(n+r)(2n+2r-2+1)-3\right]a_nx^{n+r} = 0$$

$$\sum_{n=0}^{\infty}(n+r)(2n+2r-1)a_nx^{n+r-1} - \sum_{n=a}^{\infty}\left[(n+r)(2n+2r-1)-3\right]a_nx^{n+r} = 0$$

To obtain a common power of x in each term, putting $n = m + 1$ in the first term, we get

$$\sum_{m=-1}^{\infty}(m+1+r)(2m+2+2r-1)a_{m+1}x^{m+r} - \sum_{n=0}^{\infty}\left[(n+r)(2n+2r-1)-3\right]x^{n+r} = 0$$

Since m is a dummy variable, replacing m by n,

$$\sum_{n=-1}^{\infty}(n+r+1)(2n+2r+1)a_{n+1}x^{n+r} - \sum_{n=0}^{\infty}\left[(n+r)(2n+2r-1)-3\right]a_n x^{n+r} = 0$$

$$r(2r-1)a_0x^{r-1} + \sum_{n=0}^{\infty}(n+r+1)(2n+2r+1)a_{n+1}x^{n+r} - \sum_{n=0}^{\infty}\left[(n+r)(2n+2r-1)-3\right]a_n x^{n+r} = 0$$

Equating the coefficient of the lowest degree term (i.e., x^{r-1}) to zero, the indicial equation is

$$a_0 r(2r-1) = 0$$

$$r = 0,\ r = \frac{1}{2} \qquad \left[\because\ a_0 \neq 0\right]$$

Equating the coefficient of x^{n+r} to zero,

$$(n+r+1)(2n+2r+1)a_{n+1} - \left\{(n+r)(2n+2r-1)-3\right\}a_n = 0, \qquad n \geq 0$$

$$a_{n+1} = \frac{(n+r)(2n+2r-1)-3}{(n+r+1)(2n+2r+1)}a_n, \qquad n \geq 0 \qquad \ldots(3)$$

For the first solution, putting $r = 0$ in Eq. (3),

$$a_{n+1} = \frac{n(2n-1)-3}{(n+1)(2n+1)}a_n, \qquad n \geq 0$$

Putting $n = 0, 1, 2, 3, \ldots$

$$a_1 = -3a_0$$

$$a_2 = \frac{(2-1)-3}{(2)(3)}a_1 = -\frac{2}{6}(-3a_0) = a_0$$

and so on.

Substituting $r = 0$ and values of $a_1, a_2, \ldots$ in Eq. (2),

$$y_1 = a_0 - 3a_0x + a_0x^2 + \cdots$$

$$= a_0(1 - 3x + x^2 + \cdots)$$

For the second solution, putting $r = \frac{1}{2}$ in Eq. (3),